What changes at scale is:

Concern	One device	A fleet
Deployment	SSH and docker run	Declarative, pull-based, survives offline nodes
Model update	Copy a file	Versioned artefact, staged rollout, rollback path
Failure	You notice	Nobody notices for three weeks
Hardware	Identical	Three generations, two vendors, mixed accelerators
Cost	Electricity	Bandwidth, and it is not small

The third row is important. In a cloud service a broken deploy pages somebody in minutes. At the edge, a model that degraded because winter lighting changed at one plant keeps producing confident, wrong output until a floor supervisor eventually mentions it to IT.

The four tiers

Most working edge platforms settle into the same shape, not by mandate but because the constraints push you there.

- Tier 1 - Device runtime: Inference engine, local buffering, health agent
- Tier 2 - Node orchestration: K3s/KubeEdge, workload lifecycle, device discovery
- Tier 3 - Fleet control: GitOps reconciliation, model registry, OTA pipeline
- Tier 4 - Aggregation: Telemetry store, retraining pipeline, dashboards

Decide which tier owns each capability. Trouble starts when inference logic creeps into tier 3, or fleet policy gets baked into a container shipping to tier 1.

The dashed line in Figure 1 is the whole article. Everything below it must keep working when the link drops. Everything above must tolerate nodes that vanish for a day and return with a backlog.

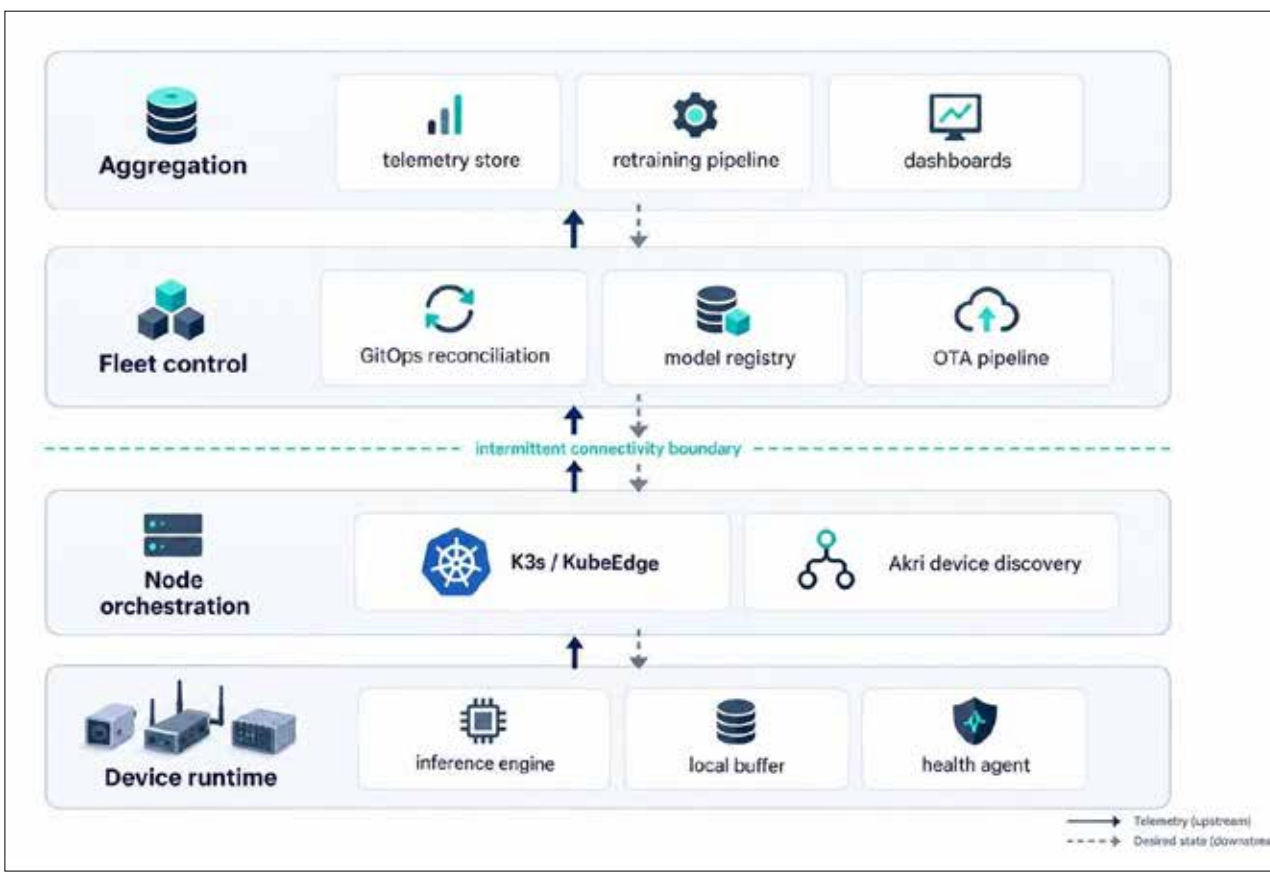


Figure 1: The four tiers, and the connectivity boundary that defines the hard problems